

Space-Time Π Identity and the Integrated Quadratic Boundary Oscillation (Route δ , Plan 2, Building Block `SPACE``TIME``IDENTITY`)

Abstract

This auxiliary block records the exact space-time radial-moment identity

$$\int_{\rho_*}^{\rho_\delta} \Pi(E_\rho, \bar{z}_\rho) d\mu(\rho) = (n-1) \int_{\tilde{\Omega}} \sigma(x) dx,$$

and isolates the quantitative centroid-route estimate

$$\int_{\rho_*}^{\rho_\delta} \Pi(E_\rho, \bar{z}_\rho) d\mu(\rho) \leq C_\Pi(n, R, \rho_*) \delta_T(\Omega). \quad (\Pi\text{-int})$$

Conditional on $(\Pi\text{-int})$, we prove the integrated quadratic boundary oscillation

$$\int_{\rho_*}^{\rho_\delta} \int_{\partial^* E_\rho} |\nu_{E_\rho} - e_{r, \bar{z}_\rho}|^2 d\mathcal{H}^{n-1} d\mu(\rho) \leq C(n, R, \rho_*) \delta_T(\Omega), \quad (\text{B}^2\text{-int})$$

and, under an additional annular/perimeter cap for the centroid slices, the integrated radial trace

$$\int_{\rho_*}^{\rho_\delta} \left(\int_{\partial^* E_\rho} |H_{\bar{z}_\rho, \rho} - 1| d\mathcal{H}^{n-1} \right)^2 d\mu(\rho) \leq C(n, R, \rho_*) \delta_T(\Omega), \quad (\text{T}_1^2\text{-int})$$

The proof of $(\text{T}_1^2\text{-int})$ uses a *slicing-then-squaring* decomposition $T_1 = T_1^{\text{rad, slic}} + T_1^{\text{tan}}$, in which the radial piece is bounded *linearly* (not at the FMP rate $\sqrt{\delta}$) via the good-ray equality of slicewise volumes and the bad-ray multiplicity bound of [1]; the squared volume Fraenkel $|E_\rho \Delta B_\rho|^2$ is controlled at rate δ via the new pointwise slice-rearrangement inequality $|E_\rho \Delta B_\rho|^2 \leq C \Pi(E_\rho, \bar{z}_\rho)$. The repaired downstream metric route in [4] no longer depends on $(\Pi\text{-int})$, but this block remains a useful diagnostic for the centroid-based space-time approach. The invalid shortcut from the rearranged Talenti profile to $(\Pi\text{-int})$ has been removed; proving $(\Pi\text{-int})$ would be an additional structural theorem, not a prerequisite for the current metric closure.

1 Notation and statement of the targets

1.1 Standing hypotheses

Fix $n \geq 2$, $R \geq 2$, $\rho_* \in (0, 1)$. Let $\Omega \subset B_R$ open with $|\Omega| = \omega_n$, $\delta := \delta_T(\Omega) \leq \delta_0(n, R, \rho_*)$. Let $u = u_\Omega$ be the variational torsion function, $t : [\rho_*, 1] \rightarrow (0, \|u\|_\infty)$ the volume-radius parametrisation, $E_\rho := \{u > t(\rho)\}$, $t_B(\rho) := (1 - \rho^2)/(2n)$, $\psi(\rho) := (-t'_B(\rho)) - (-t'(\rho)) = \rho/n + t'(\rho) \geq 0$, $\rho_\delta := 1 - \kappa\sqrt{\delta}$, $\mu(d\rho) := (-t'(\rho))d\rho$. By coarea applied to $u \in W_0^{1,2}$, E_ρ has finite perimeter for a.e. ρ .

For each ρ , $\bar{z}_\rho = \bar{z}_\rho$ is the centroid of E_ρ ([3] Theorem F). Since $E_\rho \subset B_R$, also $\bar{z}_\rho \in B_R$. No Cicalese–Leonardi containment is assumed at this stage; any lower bound on $|x - \bar{z}_\rho|$ must be justified separately on good levels or by truncating the kernel.

The L^2 divergence identity in [1]:

$$\int_{\partial^* E_\rho} |\nu_{E_\rho} - e_{r, \bar{z}_\rho}|^2 d\mathcal{H}^{n-1} = 2(P(E_\rho) - P(B_\rho)) + 2\Pi(E_\rho, \bar{z}_\rho), \quad (1)$$

where

$$\Pi(E_\rho, \bar{z}_\rho) := (n-1) \int_{B_\rho(\bar{z}_\rho) \setminus E_\rho} |x - \bar{z}_\rho|^{-1} dx - (n-1) \int_{E_\rho \setminus B_\rho(\bar{z}_\rho)} |x - \bar{z}_\rho|^{-1} dx \geq 0. \quad (2)$$

1.2 The three target theorems

Theorem 1.1 (Space-time Π identity, (G1)). $\int_{\rho_*}^{\rho_\delta} \Pi(E_\rho, \bar{z}_\rho) d\mu(\rho) = (n-1) \int_{\tilde{\Omega}} \sigma(x) dx$ for a signed Fubini density σ satisfying the kernel-weighted bound

$$|\sigma(x)| \leq \int_{J_x} |x - \bar{z}_\rho|^{-1} d\mu(\rho),$$

where $J_x := \{\rho : x \in B_\rho(\bar{z}_\rho) \Delta E_\rho\}$.

Theorem 1.2 (Integrated (B²-int) from (Π -int)). Assume (Π -int):

$$\int_{\rho_*}^{\rho_\delta} \Pi(E_\rho, \bar{z}_\rho) d\mu(\rho) \leq C_\Pi(n, R, \rho_*) \delta_T(\Omega).$$

Then

$$\int_{\rho_*}^{\rho_\delta} \int_{\partial^* E_\rho} |\nu_{E_\rho} - e_{r, \bar{z}_\rho}|^2 d\mathcal{H}^{n-1} d\mu(\rho) \leq C_{B^2}(n, R, \rho_*) \delta_T(\Omega).$$

Theorem 1.3 (Auxiliary integrated radial trace, (T_1^2 -int)). Assume (Π -int) and the auxiliary annular/perimeter cap

$$\rho_*/2 \leq |x - \bar{z}_\rho| \leq 2R \text{ on the relevant boundary pieces,} \quad P(E_\rho) - P(B_\rho) + \Pi(E_\rho, \bar{z}_\rho) \leq M_A(n, R, \rho_*)$$

for a.e. $\rho \in [\rho_*, \rho_\delta]$. Then

$$\int_{\rho_*}^{\rho_\delta} \left(\int_{\partial^* E_\rho} |H_{\bar{z}_\rho, \rho} - 1| d\mathcal{H}^{n-1} \right)^2 d\mu(\rho) \leq C_{\text{radial}}(n, R, \rho_*) \delta_T(\Omega).$$

2 Space-time identity (G1)

By (2), $\Pi(E_\rho, \bar{z}_\rho) = (n-1) \int_{\mathbb{R}^n} (\mathbf{1}_{B_\rho(\bar{z}_\rho)} - \mathbf{1}_{E_\rho}) |x - \bar{z}_\rho|^{-1} dx$. By joint Borel measurability and Fubini,

$$\int_{\rho_*}^{\rho_\delta} \Pi(E_\rho, \bar{z}_\rho) d\mu(\rho) = (n-1) \int_{\mathbb{R}^n} \sigma(x) dx, \quad (3)$$

where $\sigma(x) := \int_{\rho_*}^{\rho_\delta} (\mathbf{1}_{B_\rho(\bar{z}_\rho)} - \mathbf{1}_{E_\rho}) / |x - \bar{z}_\rho| d\mu(\rho)$. The integrand vanishes outside $\tilde{\Omega} := B_{R+1}$ (which contains $\Omega \subset B_R$), because $\bar{z}_\rho \in B_R$ and $\rho \leq 1$. The kernel is locally integrable in dimension $n \geq 2$; if desired, one first proves the identity with $\min\{|x - \bar{z}_\rho|^{-1}, M\}$ and then lets $M \rightarrow \infty$ by monotone convergence applied to the positive and negative parts. Taking absolute values inside the ρ -integral gives the stated kernel-weighted crossing bound. $\square$

3 The remaining quantitative input

The identity of Theorem 1.1 rewrites $\int \Pi(E_\rho, \bar{z}_\rho) d\mu$ as an integral over the space–time mismatch

$$J_x := \{\rho \in [\rho_*, \rho_\delta] : x \in B_\rho(\bar{z}_\rho) \Delta E_\rho\}.$$

The rearranged Talenti/Nicola–Tilli profile gap controls the one-dimensional distributional profiles $t(\rho)$ and $t_B(\rho)$. It does *not*, by itself, control the physical crossing set J_x , because J_x also depends on the moving spatial radius $|x - \bar{z}_\rho|$. Therefore the implication

$$\mathcal{L}^1(J_x) \lesssim \delta_T(\Omega)$$

is not asserted here. The additional theorem that would finish this centroid subroute is the following integrated moment estimate.

Theorem 3.1 (Integrated Π -control; remaining centroid-route input). *There is $C_\Pi(n, R, \rho_*) > 0$ such that, for every bounded open $\Omega \subset B_R$ with $|\Omega| = \omega_n$ in the small-deficit regime,*

$$\int_{\rho_*}^{\rho_\delta} \Pi(E_\rho, \bar{z}_\rho) d\mu(\rho) \leq C_\Pi(n, R, \rho_*) \delta_T(\Omega). \quad (\Pi\text{-int})$$

Remark 3.2 (Possible proof route). A viable proof of Theorem 3.1 would have to combine the Fubini identity of Theorem 1.1 with a genuine space–time coarea or transport estimate for the physical crossings of $|x - \bar{z}_\rho| = \rho$. The H^1 -in- ρ centroid estimate of [3] is the natural input for controlling the moving centre term, but an extra argument is still needed to convert the profile gap into the physical crossing-length bound.

4 Assembly of (B²-int) from (Π -int)

Assume (Π -int). By (1) integrated,

$$\int |\nu - e_{r, \bar{z}_\rho}|^2 dH d\mu = 2 \int (P - P_\rho) d\mu + 2 \int \Pi(E_\rho, \bar{z}_\rho) d\mu.$$

The first piece is $\leq C\delta$ via the deficit identity in [1]. The second piece is exactly (Π -int). Both pieces are $O(\delta)$, giving Theorem 1.2. $\square$

5 The integrated radial trace (T_1^2 -int) — slicing-then-squaring

Write $T_1 = T_1(E_\rho, \bar{z}_\rho) := \int_{\partial^* E_\rho} |x - \bar{z}_\rho|/\rho - 1| d\mathcal{H}^{n-1}$ and decompose

$$T_1 = T_1^{\text{rad, slic}} + T_1^{\text{tan}},$$

with

$$\begin{aligned} T_1^{\text{rad, slic}} &:= \int_{\partial^* E_\rho} ||x - \bar{z}_\rho|/\rho - 1| \cdot |e_{r, \bar{z}_\rho} \cdot \nu_{E_\rho}| d\mathcal{H}^{n-1}, \\ T_1^{\text{tan}} &:= \int_{\partial^* E_\rho} ||x - \bar{z}_\rho|/\rho - 1| \cdot (1 - |e_{r, \bar{z}_\rho} \cdot \nu_{E_\rho}|) d\mathcal{H}^{n-1}. \end{aligned}$$

By the slicing identity [6, Thm 18.11] applied with $g(r) := |r/\rho - 1|$:

$$T_1^{\text{rad, slic}} = \int_{S^{n-1}} \sum_{j=1}^{N(\theta)} |r_{\theta, j}/\rho - 1| r_{\theta, j}^{n-1} d\theta. \quad (4)$$

5.1 Pointwise bound for $T_1^{\text{rad,slic}}$

Lemma 5.1. $T_1^{\text{rad,slic}} \leq C_4(n, R, \rho_*) [|E_\rho \Delta B_\rho(\bar{z}_\rho)| + (P(E_\rho) - P(B_\rho))].$

Proof. Good rays. On a good ray ($N = 1$), the slicewise 1D symmetric-difference equals:

$$\int_0^\infty |\mathbf{1}_{E_{\rho,\theta}} - \mathbf{1}_{(0,\rho)}| r^{n-1} dr = \frac{1}{n} |r_{\theta,1}^n - \rho^n|. \quad (5)$$

By MVT, $|r_{\theta,1}/\rho - 1| \cdot r_{\theta,1}^{n-1} \leq \beta_n |r_{\theta,1}^n - \rho^n|$ with $\beta_n = \beta_n(R, \rho_*)$. Integrating and using spherical Fubini:

$$\int_{\{N=1\}} |r_{\theta,1}/\rho - 1| r_{\theta,1}^{n-1} d\theta \leq n\beta_n |E_\rho \Delta B_\rho(\bar{z}_\rho)|.$$

Bad rays. $|r_{\theta,j}/\rho - 1| r_{\theta,j}^{n-1} \leq C(R, \rho_*)$, summed gives $\leq (2R)^{n-1}(1 + k(\theta))$. We claim the bad-ray multiplicity bound

$$\int_{S^{n-1}} k(\theta) d\theta \leq (\rho_*/2)^{-(n-1)} (P(E_\rho) - P(B_\rho)),$$

which we prove as follows (note: this estimate is conditional on the auxiliary annular cap of Theorem 1.3, which provides $|x - \bar{z}_\rho| \geq \rho_*/2$ on the relevant boundary pieces). For each direction $\theta \in S^{n-1}$, let $k(\theta) \geq 0$ count the crossings of the radial ray with $\partial^* E_\rho$ beyond the spherical baseline. By the radial-projection inequality and the area formula, $\int_{S^{n-1}} \sum_j r_{\theta,j}^{n-1} d\theta = P_{\text{rad}}(E_\rho) \leq P(E_\rho)$, where $r_{\theta,j}$ are the crossing radii. Each crossing contributes $\geq (\rho_*/2)^{n-1}$ to this integral, since $|x - \bar{z}_\rho| \geq \rho_*/2$ on the annular cap. Summing the contributions beyond the spherical baseline gives $\int_{S^{n-1}} k(\theta) d\theta \leq (\rho_*/2)^{-(n-1)} (P(E_\rho) - P(B_\rho))$, as claimed.

Combining the two estimates gives the claim. $\square$

5.2 Pointwise bound for T_1^{tan}

Lemma 5.2. $T_1^{\text{tan}} \leq (P(E_\rho) - P(B_\rho)) + \Pi(E_\rho, \bar{z}_\rho).$

Proof. Under the auxiliary annular cap, $\|x - \bar{z}_\rho\|/\rho - 1 \leq C(R, \rho_*)$, so $T_1^{\text{tan}} \leq C(R, \rho_*) \int (1 - |e_{r,\bar{z}_\rho} \cdot \nu|) dH \leq C(R, \rho_*) \int (1 - e_{r,\bar{z}_\rho} \cdot \nu) dH = C(R, \rho_*) ((P - P_\rho) + \Pi(E_\rho, \bar{z}_\rho))$ by the divergence identity in [1]. $\square$

5.3 Squaring and integrating

By Lemmas 5.1, 5.2 and $(a + b + c)^2 \leq 3(a^2 + b^2 + c^2)$:

$$T_1^2 \leq 3C_5^2 [|E_\rho \Delta B_\rho|^2 + (P - P_\rho)^2 + \Pi(E_\rho, \bar{z}_\rho)^2].$$

Integrating against $d\mu$ and bounding each piece by $C\delta$:

- $\int |E_\rho \Delta B_\rho|^2 d\mu \leq C_8 \int \Pi(E_\rho, \bar{z}_\rho) d\mu \leq C\delta$ via the slice-rearrangement (§6) + (G3).
- $\int (P - P_\rho)^2 d\mu \leq M_A \int (P - P_\rho) d\mu \leq C\delta$ under the auxiliary cap (uses $(P - P_\rho) \geq 0$ to bound $(P - P_\rho)^2 \leq M_A(P - P_\rho)$ on the cap).
- $\int \Pi(E_\rho, \bar{z}_\rho)^2 d\mu \leq M_A \int \Pi(E_\rho, \bar{z}_\rho) d\mu \leq C\delta$ under the same cap.

Hence $\int T_1^2 d\mu \leq C_{T_1^2} \delta$.

Combining with (G3) gives $(T_1^2\text{-int})$ as follows. Since $H_{\bar{z}_\rho, \rho} = (|x - \bar{z}_\rho|/\rho)(e_{r,\bar{z}_\rho} \cdot \nu)$, we decompose

$$H_{\bar{z}_\rho, \rho} - 1 = (|x - \bar{z}_\rho|/\rho - 1)(e_{r,\bar{z}_\rho} \cdot \nu) + (e_{r,\bar{z}_\rho} \cdot \nu - 1),$$

where on the annular cap $|e_{r,\bar{z}_\rho} \cdot \nu| \leq 1$ and $||x - \bar{z}_\rho|/\rho - 1| \leq C(R, \rho_*)$, so both factors are bounded. Taking absolute values and integrating over $\partial^* E_\rho$ yields

$$\int_{\partial^* E_\rho} |H_{\bar{z}_\rho, \rho} - 1| d\mathcal{H}^{n-1} \leq T_1 + \int_{\partial^* E_\rho} |e_{r,\bar{z}_\rho} \cdot \nu - 1| d\mathcal{H}^{n-1}.$$

For the second term, using $|e_{r,\bar{z}_\rho} \cdot \nu - 1| \leq |e_{r,\bar{z}_\rho} - \nu|$ and applying Cauchy–Schwarz with respect to $d\mathcal{H}^{n-1} d\mu$ together with the perimeter bound $P(E_\rho) \leq M_A$ on the bounded reduction,

$$\left(\int |e_{r,\bar{z}_\rho} - \nu| d\mathcal{H}^{n-1} d\mu \right)^2 \leq M_A \cdot \int |e_{r,\bar{z}_\rho} - \nu|^2 d\mathcal{H}^{n-1} d\mu.$$

The L^2 -bound on $|e_{r,\bar{z}_\rho} - \nu|^2$ provided by (G3) (Theorem 1.2) then gives an $O(\delta)$ control. Squaring the pointwise decomposition, applying $(a+b)^2 \leq 2(a^2 + b^2)$, integrating against $d\mu$, and combining with $\int T_1^2 d\mu \leq C_{T_1^2} \delta$ yields (T₁-int). $\square$

6 The slice-rearrangement inequality

Lemma 6.1. *Under the annular cap $\rho_*/2 \leq r \leq 2R$, $|E_\rho \Delta B_\rho(\bar{z}_\rho)|^2 \leq C_8(n, R, \rho_*) \Pi(E_\rho, \bar{z}_\rho)$ pointwise in the smallness regime.*

Proof. Step 1: $\Pi(E_\rho, \bar{z}_\rho) \geq cM_1$. Let $M_1 := \int_{E_\rho \Delta B_\rho} |\rho - |x - \bar{z}_\rho|| dx$. By rewriting (2),

$$\Pi(E_\rho, \bar{z}_\rho) = (n-1) \int_{E_\rho \Delta B_\rho} \frac{|\rho - |x - \bar{z}_\rho||}{\rho |x - \bar{z}_\rho|} dx \geq \frac{n-1}{2R\rho} M_1.$$

Step 2: $|E_\rho \Delta B_\rho|^2 \leq C M_1$. By polar coordinates and a slicewise Cauchy–Schwarz on $[\rho_*/2, 2R]$, $\sigma(\theta)^2 \leq 4R m_1(\theta)$ with $\sigma(\theta) = \mathcal{L}^1(E_{\rho,\theta} \Delta (0, \rho))$, $m_1(\theta) = \int |r - \rho| \mathbf{1}_{E_{\rho,\theta} \Delta (0, \rho)} dr$. Cauchy–Schwarz on S^{n-1} then gives $(\int \sigma d\theta)^2 \leq 4R |S^{n-1}| \int m_1 d\theta$. Since $\sigma(\theta) d\theta \cdot r^{n-1}$ (for $\rho_*/2 \leq r \leq 2R$) gives the volume element up to the comparable factors $(\rho_*/2)^{n-1}$ and $(2R)^{n-1}$, the slice-to-volume conversion preserves the bound up to constants depending only on n, ρ_*, R ; this yields $|E_\rho \Delta B_\rho|^2 \leq C_9 M_1$.

Combining Steps 1–2 gives the claim. $\square$

7 Discussion: why T_1 -route closes where T_2 -route stalls

Remark 7.1. The T_2 -route would extract on good rays $(r_{\theta,1}/\rho - 1)^2 r_{\theta,1}^{n-1} \leq C |r_{\theta,1}^{n-1} - \rho^{n-1}|$ (a square on LHS, unsigned mismatch on RHS), then need $\int |r_{\theta,1}^{n-1} - \rho^{n-1}| d\theta \leq C\sqrt{\delta_T}$ (FMP rate). Integrating against $d\mu$ leaks at $\sqrt{\delta}$.

The T_1 -route uses linear good-ray equality (5) and the slice-rearrangement to get $|E_\rho \Delta B_\rho|^2 \leq C\Pi$, which integrates to $C\delta$ via (G3). Squaring *before* integrating avoids the FMP bottleneck.

8 Conditionality

Inputs used after (II-int) and the auxiliary annular/perimeter cap: slicing identity (Maggi); bad-ray multiplicity bound (elementary, see proof of Lemma 5.1); slicewise good-ray equality; spherical Fubini; divergence identity (5.3); Talenti profile gap; slice-rearrangement Lemma 6.1; (F) of [3].

Inputs explicitly not used: the per- ρ pointwise radial-trace inputs (1.1), (1.2)=(R), (2.4); pointwise (B²); the boundary-layer pointwise (7.1) of [5]; any selection, graph entry, Schauder, Fuglede.

Centroid-route status: Theorem 3.1, plus the annular/perimeter cap stated in Theorem 1.3, would complete this centroid subroute. Without them the present block proves the exact space–time identity and a conditional centroid closure only. The repaired metric closure is instead supplied by [4].

Remark 8.1 (How the metric closure uses D_H and D_I jointly). The repaired metric route [4] bypasses Theorem 3.1 (and hence the centroid space–time control of physical crossings) by pairing the two pieces of the deficit identity $D_H + D_I$ of [1] as two independent integral inputs:

- the integrated isoperimetric defect $\int (P(E_\rho) - P(B_\rho))(P(E_\rho) + P(B_\rho)) d\mu \leq C\delta_T$ (i.e. $\int D_I d\mu \leq C\delta_T$) controls the $L^2(d\mu)$ -norm of the levelwise Fraenkel distance via Fusco–Julin [4, Lemma 3.1];
- the integrated Cauchy–Schwarz defect $\int D_H d\mu \leq C\delta_T$ controls the $L^2(d\mu)$ -norm of the metric derivative $|\dot{F}_\rho|_{\mathcal{X}}$ in the quotient space $\mathcal{X}$ of [2] via its metric first variation.

The first input says “many levels are close to balls” but does not fix *which* level; the second says “the family $\rho \mapsto E_\rho$ does not curve away from its trajectory faster than $\sqrt{\delta_T}$ in L^2 ”. Combining them on the order-one window $J^* = [\rho_\delta - L^*, \rho_\delta]$ via Markov (on the distance term) + Cauchy–Schwarz kinetic transport (on the velocity term) yields a *uniform* pointwise bound $d_{\mathcal{X}}(F_\rho, B_1)^2 \leq C_*\delta_T$ on J^* , including the endpoint $\rho = \rho_\delta$ where the boundary-layer transfer of [5] has minimal thickness $1 - \rho_\delta \asymp \sqrt{\delta_T}$. This is the D_H/D_I -route to sharp Faber–Krahn stability: each integrated defect supplies a different one of the two H^1 -in- ρ ingredients required by the 1D Sobolev embedding underlying the pointwise endpoint bound, while neither alone would suffice. The centroid space–time Π -control of Theorem 3.1 is logically independent of this argument and would only be needed for the alternative centroid-route closure.

References

- [1] Plan 2 building block LEVELSETIDENTITY.
- [2] Plan 2 building block METRICFRAMEWORK.
- [3] Plan 2 building block CENTROIDBOUND.
- [4] Plan 2 building block WEIGHTEDMETRICTRACE.
- [5] Plan 2 building block BOUNDARYLAYERTRANSFER.
- [6] F. Maggi, Cambridge Univ. Press, 2012. Thm 15.9, 18.11, 19.4.